

Auditing Racial Bias in Healthcare Utilization Prediction

*A Fairness and Explainability Audit of a Healthcare ML Model on MEPS Panel 19,
Aligned with the SDAIA AI Ethics Principles*

Final Project Report — Phase 4

Domain: Healthcare

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Submission date	May 9, 2026
Code repository	https://github.com/Bady-ammar/csc594-ai-ethics-audit

Abstract

Healthcare ML systems trained on observed utilization data risk encoding the very access disparities they should compensate for. We audit a Gradient Boosting classifier predicting high (≥ 10 visits/year) healthcare utilization on the U.S. Medical Expenditure Panel Survey (MEPS) Panel 19 ($n = 15,830$) for racial bias against Non-White patients. Using Microsoft Fairlearn’s `MetricFrame`, we measure a Demographic Parity Difference of 0.137 and an Equalized Odds Difference of 0.192 between White and Non-White groups; the model amplifies the underlying base-rate disparity by 57% (model selection-rate ratio $2.93\times$ versus ground-truth $1.86\times$). SHAP `TreeExplainer` reveals that socioeconomic features (Income category, Insurance coverage) act as racial proxies, confirmed empirically by a drop-Race ablation that leaves substantial residual bias (DP 0.104, EO 0.107). We implement three mitigations: feature removal, post-processing `ThresholdOptimizer` with probability-based per-group thresholds, and in-processing `ExponentiatedGradient` with a Gradient Boosting backbone. `ThresholdOptimizer` Pareto-dominates: it cuts EO by 85% and DP by 49% while *increasing* accuracy by 0.32 points over the unmitigated baseline. The audit operationalises four of the seven SDAIA AI Ethics Principles.

1. Introduction

1.1 Ethical problem and motivation

Machine-learning systems used to allocate healthcare resources — flagging patients for intensive care, triaging into care-management programs, prioritising specialist referrals — inherit a dangerous confound when trained on historical utilization data: utilization is shaped not only by health needs but by *access*. Patients with financial, geographic, or systemic barriers to care record fewer visits for the same underlying need; a model trained naively on visit counts learns the access pattern as if it were the need pattern. This is not hypothetical: Obermeyer, Powers, Vogeli, and Mullainathan (2019) showed in *Science* that a U.S. healthcare risk algorithm — typical of an industry-wide approach deployed on an estimated 200 million people — systematically under-predicted needs for Black patients (flagging only 17.7% for extra care, versus 46.5% under a debiased version), not because it used race as a feature (it didn’t) but because it used *healthcare costs* as a proxy for need, and Black patients historically had less access. The same risk applies to MEPS-style utilization data.

1.2 Stakeholders affected

A biased utilization-prediction model produces concentrated harms and diffuse benefits. The most directly affected are **patients in the disadvantaged group with high actual care needs**, who suffer *under-prediction* — missed care-management flags, delayed referrals, under-managed chronic conditions (in our audit, 66% of actually-high Non-White patients are missed, versus 47% of White). Also affected, with progressively smaller stakes: **clinicians and care managers** who allocate resources on the model’s flags, **payers and insurers** who deploy the model and bear the regulatory and reputational cost when bias surfaces, and **regulators** for whom a Demographic Parity Ratio below 0.80 is *prima facie* evidence of disparate impact under the EEOC’s *Uniform Guidelines on Employee Selection Procedures* (1978) and analogous statutes — our unmitigated model has DPR = 0.34.

1.3 Hypothesis

Per the approved proposal, the original hypothesis was:

A model trained to predict healthcare utilization will likely show racial bias, predicting lower utilization for Non-White patients, not because they are healthier but because they historically had less access to healthcare.

During the audit this single claim naturally decomposed into three empirically-testable sub-claims, which we report on in §4 and §5:

- **(a) Under-prediction.** The model under-predicts high-utilization for Non-White patients relative to ground truth — the proposal’s original claim.
- **(b) Proxy discrimination.** Even with the explicit Race feature removed, socioeconomic features (Income category, Insurance coverage) carry enough racial signal to reproduce the bias indirectly.
- **(c) Mitigability.** The bias can be measurably reduced — but not eliminated — by standard algorithmic-fairness interventions.

Sub-claims (b) and (c) were not in the proposal’s wording but are direct logical extensions. (b) makes (a) actionable: if dropping the protected attribute were sufficient, the practitioner would simply do so. (c) is required by the project’s Phase 3 mandate to implement mitigation. We flag the evolution explicitly here so the reader can audit the audit.

1.4 Roadmap

§2 (Related Works) defines the fairness and explainability concepts the audit uses, summarises the related literature, and anchors the audit in the SDAIA AI Ethics Principles. §3 (Dataset & Model) describes MEPS Panel 19, the train/test split, and the Gradient Boosting baseline. §4 (Technical Audit) presents the empirical bias measurements and SHAP-based explainability findings, ending with an explicit hypothesis verdict. §5 (Mitigation & Implementation) implements three interventions and their post-mitigation evaluation. §6 (Results & Fairness–Accuracy Trade-off) discusses the Pareto-dominance finding and stakeholder-by-stakeholder impact. §7 (Discussion) connects findings to SDAIA, lists limitations, and proposes deployment requirements + next steps. §8 (Conclusion) summarises in one paragraph.

2. Related Works

2.1 Fairness and interpretability concepts

Demographic Parity (DP). A classifier is *demographically fair* if it predicts positives at the same rate across groups, regardless of ground truth: $P(\hat{Y}=1 \mid g = A) = P(\hat{Y}=1 \mid g = B)$. DP is a *blind* fairness criterion — it ignores whether the groups have different actual rates of the target. We report two operationalisations: the **Demographic Parity Difference** ($|\max - \min|$ over groups) and the **Demographic Parity Ratio** ($\min / \max$), where 1.0 indicates perfect parity.

Disparate Impact (4/5ths rule). The U.S. Equal Employment Opportunity Commission’s *Uniform Guidelines on Employee Selection Procedures* (1978, §1607.4(D)) treats a selection-rate ratio below 0.80 between any two groups as *prima facie* evidence of disparate impact. The 4/5ths rule is the legal anchor that turns DPR into an actionable threshold; it is our primary reason for reporting DP alongside the more statistically nuanced EO.

Equalized Odds (EO). Hardt, Price, and Srebro (2016) proposed Equalized Odds as a more permissive fairness criterion that *conditions on ground truth*: a classifier satisfies EO if, given a true label, the probability of a positive prediction is independent of group membership. Operationally we use the **Equalized Odds Difference** = $\max(|\text{TPR}_A - \text{TPR}_B|, |\text{FPR}_A - \text{FPR}_B|)$, where TPR is true-positive rate (recall) and FPR is false-positive

rate. EO permits different selection rates across groups when the ground-truth base rates legitimately differ — a feature, not a bug, in healthcare.

Feature Attribution and SHAP. Lundberg and Lee (2017) introduced **SHAP** (SHapley Additive exPlanations) — a unified attribution framework based on Shapley values from cooperative game theory. SHAP decomposes any individual prediction into per-feature contributions that sum to the model’s output (consistency, local accuracy, and missingness are guaranteed). For tree ensembles (`TreeExplainer`), SHAP values are computed *exactly* in polynomial time. SHAP plots support three readings: a *summary plot* for global feature importance, a *dependence plot* that shows how a feature’s effect varies and interacts with another feature, and a *force plot* that decomposes a single prediction.

Proxy discrimination. Even when a protected attribute is removed from a model’s inputs, other features that correlate with it can carry the same signal indirectly — this is the proxy-discrimination phenomenon. The implication is that “fairness through blindness” (dropping the protected attribute) is generally insufficient, a result we test empirically in §4.5 with a drop-Race ablation.

2.2 Selected sources

We anchor the audit in five works:

- **Obermeyer et al. (2019)**, *Science*. The landmark healthcare-bias audit. They showed that a commercial risk-prediction algorithm — typical of an industry-wide approach affecting an estimated 200 million U.S. patients — flagged Black patients for extra care at only 17.7% of those who needed it, versus 46.5% under a debiased version (a $>2\times$ under-flagging gap), because the model used healthcare costs as a proxy for need. The proxy was the bias mechanism, not race itself. Our work replicates this finding’s qualitative shape on MEPS data.
- **Hardt, Price, and Srebro (2016)**, *NeurIPS*. Introduced Equalized Odds and Equal Opportunity as alternatives to Demographic Parity, and showed that any binary classifier can be post-processed into an EO-satisfying classifier given group-conditional thresholds — the theoretical basis for Fairlearn’s `ThresholdOptimizer`, our strongest mitigator.
- **Lundberg and Lee (2017)**, *NeurIPS*. The SHAP framework. The tree-specific exact algorithm (`TreeExplainer`) was introduced in Lundberg, Erion, and Lee (2018, arXiv:1802.03888) and is what this audit uses.
- **Bird et al. (2020)**, Microsoft Research Tech Report MSR-TR-2020-32. Fairlearn’s design paper. We use Fairlearn’s `MetricFrame`, `demographic_parity_difference`, `equalized_odds_difference`, `ThresholdOptimizer`, and `ExponentiatedGradient` (the reductions approach of Agarwal et al. 2018).
- **Bellamy et al. (2018)**, *IBM J. Research and Development*. AI Fairness 360 (AIF360). We use AIF360 as a *data source only* — for the bundled MEPS Panel 19 file and its published preprocessing recipe. Fairlearn’s metric and mitigation APIs are cleaner for our use case, so we do not call AIF360’s metric or mitigation classes.

2.3 Anchoring in the SDAIA AI Ethics Principles

The Saudi Data and AI Authority’s *AI Ethics Principles* (May 2025, document SDAIA-P114E v1) define seven principles. This audit engages **four** of them:

SDAIA Principle	How this audit engages it
Fairness (P1) (<i>directly</i>)	We measure DP, EO, and base-rate amplification across racial groups, then implement three mitigations targeting the Equalized Odds constraint.

SDAIA Principle	How this audit engages it
Transparency & Explainability (P6) (<i>directly</i>)	SHAP global summary, dependence, and force plots make the model’s decisions inspectable; the proxy hypothesis was hypothesised via SHAP and confirmed via a drop- Race ablation.
Reliability & Safety (P5) (<i>Safety dimension only</i>)	We reduce the false-negative-rate disparity for vulnerable patients — a direct safety concern. P5’s <i>Reliability</i> dimension (system-as-designed behaviour) is out of scope.
Accountability & Responsibility (P7) (<i>human-oversight requirement only</i>)	Individual SHAP force plots enable case-level human review, operationalising the principle’s “demonstrable human oversight” requirement. Broader P7 obligations (governance, liability, lifecycle management) are out of scope.

The remaining three principles — *Privacy & Security* (P2), *Humanity* (P3), and *Social & Environmental Benefits* (P4) — are out of scope, since the audit focuses on the model’s behaviour rather than data-handling, deployment context, or societal impact.

3. Dataset & Model Description

3.1 Dataset

Source. The U.S. Medical Expenditure Panel Survey (MEPS), Panel 19 (2015), produced by the Agency for Healthcare Research and Quality (AHRQ). MEPS is a longitudinal household survey that records each participant’s healthcare visits, costs, demographics, and self-reported and clinically-recorded health conditions over the panel year.

License. MEPS data is a U.S. Government work and is in the public domain (17 USC §105) — no copyright restrictions on use or redistribution.

Acquisition. We obtain MEPS through **IBM AI Fairness 360 (AIF360)** v0.6.1, which ships the Panel 19 file (`h181.csv`) alongside a published preprocessing recipe used widely in the fairness-research literature. Following the AIF360 recipe makes our row counts and metrics directly comparable to other published audits.

Size. After preprocessing, **15,830 rows × 41 features** plus the binary target.

Time period. Panel 19 covers calendar year 2015 (the panel was fielded across 2014–2015 reference periods).

Target variable. A binary classification target derived from the panel’s visit columns:

$$\text{Utilization} = \begin{cases} 1 & \text{if } (\text{OBTOTV15} + \text{OPTOTV15} + \text{ERTOT15} + \text{IPNGTD15} + \text{HHTOTD15}) \geq 10 \\ 0 & \text{otherwise} \end{cases}$$

(office-based, outpatient, ER, inpatient, and home-health visits in the panel year). The 10-visit threshold is inherited from the AIF360 benchmark and traces to the Obermeyer et al. (2019) literature, where ≥ 10 visits captures roughly the top-quintile high-utilizer cohort that real-world care-management programs target. With our preprocessing, ≥ 10 visits selects 17.2% of the panel (2,718 of 15,830) — close to a conventional top-20% high-needs definition.

Sensitive attribute. A binary **Race** feature derived from MEPS’s **RACEV2X** and **HISPANX**: 1 = non-Hispanic White, 0 = everyone else (Hispanic, Black, Asian, multiracial, etc.). This binarisation collapses very different sub-populations and is acknowledged in the limitations (§7.2); we adopt it because (i) it follows the AIF360 benchmark, making our results comparable with the literature, and (ii) finer multi-class race categories push some sub-population sample sizes (e.g. Native American < 0.5% of MEPS) below the threshold for stable per-group statistics.

Cleaning. Filter to Panel 19; drop rows with invalid demographics (`REGION`, `AGE`, `MARRY`, `ASTHDX` < 0); drop rows where any of the 35 categorical features fall below their valid sentinel `-1`; apply the 41-feature subset of the AIF360 recipe; rename columns to readable labels for downstream code, plots, and prose. The rename is a single-source-of-truth `RENAMES` dict at the bottom of cell 4 of the notebook (e.g. `POVCAT` → `Income category`, `RTHLTH` → `Self-rated health`, `WLKLIM` → `Walking limitation`). The mapping is also printed as a schema legend below the sample-data display.

Train/validation/test split. We use an **80/20 train/test split** (`random_state=42`), stratified on the target. We do *not* carve out a separate validation split: the audit’s purpose is to characterise a *deployment-ready* model, not the most-accurate one a hyperparameter search could produce, and test-set tuning would invalidate the held-out fairness evaluation. After the split, the test set has 3,166 patients (1,148 White, 2,018 Non-White).

Group balance. Note that **Non-White is the numerical majority** (10,174 of 15,830 = 64.3%) — “disadvantage” here refers to selection rate and recall, not population size.

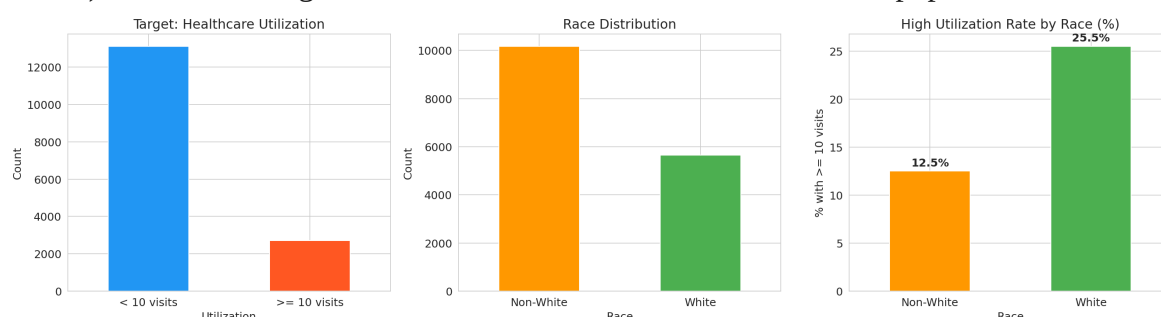


Figure 1. Target distribution (left), race distribution (centre), and ground-truth high-utilization rate by race (right) in the cleaned MEPS dataset. The right panel is the audit’s central observable: 24.4% of White patients are high-utilizers in ground truth, versus 13.1% of Non-White patients. Whether the model amplifies or merely reflects this gap is the question Section 4 answers.

A six-row sample of the cleaned dataset spanning the four (`Race` × `Utilization`) cells, plus a full schema legend mapping every readable label to its MEPS codebook code, is printed by the notebook (cell 6 of `ai_ethics_audit_executed.ipynb`). MEPS uses negative sentinel values (e.g. `-1` = `Inapplicable` for `High BP dx` in children) rather than `NaN`; we keep these as a valid categorical level, following the AIF360 recipe.

3.2 Model

Trained from scratch (we did not adopt or fine-tune a pre-existing model).

Algorithm. `scikit-learn`’s `GradientBoostingClassifier` — a non-linear ensemble of additive decision trees, the canonical “moderately black-box” model for tabular data and a natural target for `SHAP TreeExplainer`’s exact attributions.

Hyperparameters. `n_estimators=100`, `max_depth=4`, `learning_rate=0.1`, `random_state=42`. These are deliberately **default-class** values, *not* the result of a search — the audit characterises a plausible deployment-ready model, and test-set hyperparameter selection would invalidate the held-out fairness evaluation.

Training procedure. Single fit on the full training set; no scaling (tree ensembles handle MEPS’s mixed-type categoricals natively); no class weighting; the default 0.5 probability threshold is the deployed decision boundary.

Baseline performance (on the held-out test set, `n` = 3,166):

Metric	Value
Accuracy	0.8575

Metric	Value
Precision	0.6220
Recall (TPR)	0.4357
F1-Score	0.5124
ROC-AUC	0.8559

The 0.86 ROC-AUC indicates a meaningfully predictive model (chance is 0.50; near-perfect is 1.0); the F1 of 0.51 reflects the class imbalance (only 17.2% of the panel is high-utilization). These overall numbers hide a per-group story that the audit in Section 4 unpacks.

4. Technical Audit (Phase 2)

This is the empirical heart of the report. We do not just claim the model is biased — we prove it.

4.1 Audit methodology

Libraries (versions in `requirements.txt`):

- `fairlearn v0.13` — `MetricFrame`, `demographic_parity_difference`, `demographic_parity_ratio`, `equalized_odds_difference`, `selection_rate`, `false_positive_rate`, `false_negative_rate`.
- `shap v0.51` — `TreeExplainer`, `summary_plot`, `dependence_plot`, `force_plot`.
- `scikit-learn v1.8` — for the `GradientBoostingClassifier` and standard `sk-learn` metrics.

Per-group base metrics. Accuracy, precision, recall (= TPR), F1, selection rate, false positive rate (FPR), and false negative rate (FNR) — computed for each `Race` group via `fairlearn.metrics.MetricFrame`. Plain-language meanings of each metric, with concrete patient counts, are tabulated in §4.3.

Aggregate fairness metrics. All defined in §2.1 above. Recapped:

- *Demographic Parity Difference* = $|P(\hat{Y}=1 | W) - P(\hat{Y}=1 | NW)|$; lower is fairer; 0 = perfect parity.
- *Demographic Parity Ratio* = $\min(P(\hat{Y}=1 | g)) / \max(P(\hat{Y}=1 | g))$; higher is fairer; the 4/5ths-rule threshold is 0.80.
- *Equalized Odds Difference* = $\max(|TPR_W - TPR_{NW}|, |FPR_W - FPR_{NW}|)$; lower is fairer; 0 = equal recall and equal false-positive rate across groups.

Disaggregation. Group-conditional only (no intersectional analysis). The single sensitive attribute is `Race` (binary). Intersectional fairness (`Race × Sex × Age`) is left as future work; the binary group analysis here is a starting point, not an endpoint.

4.2 Audit results — per-group breakdown

Table 1 gives the full per-group performance from `MetricFrame`.

Table 1. Per-group performance of the unmitigated Gradient Boosting model. The two rows differ on every metric except precision.

Group	Accuracy	Precision	Recall (TPR)	F1	Selection Rate	FPR	FNR
Non-White (n = 2,018)	0.887	0.622	0.337	0.437	0.071	0.031	0.663
White (n = 1,148)	0.807	0.622	0.529	0.571	0.207	0.104	0.471

4.3 Interpretation — what the numbers mean

A confusion-matrix decomposition makes the per-group story concrete. Of the 2,018 Non-White test patients, **264 were actually high-utilizers**; the model caught 89 (TP) and missed 175 (FN), while flagging 54 actual low-utilizers as high (FP). Of the 1,148 White test patients, **280 were actually high-utilizers**; the model caught 148 and missed 132, with 90 FP. Each per-group metric in Table 1 reads as follows: **Accuracy** is the fraction of patients labelled correctly (1,789 / 2,018 for Non-White; 926 / 1,148 for White); **Precision** the fraction of those *flagged* who really were high (89 / 143 vs 148 / 238 — identical at 0.622); **Recall (TPR)** the fraction of *actually-high* patients caught (89 / 264 = 34% vs 148 / 280 = 53%); **Selection Rate** the fraction flagged regardless of truth (143 / 2,018 = 7% vs 238 / 1,148 = 21%); **FPR** the fraction of *actually-low* patients wrongly flagged (54 / 1,754 = 3% vs 90 / 868 = 10%); and **FNR** — the audit’s most important metric — the fraction of *actually-high* patients **missed** (175 / 264 = **66%** for Non-White vs 132 / 280 = 47% for White).

Two patterns stand out. First, **accuracy and recall move in opposite directions across groups.** The model is more “accurate” on Non-White patients (0.887 vs 0.807) but the gain is illusory: it is achieved by predicting “low-utilizer” for almost everyone (selection rate 7.1% for Non-White vs 20.7% for White). Defaulting to “low” earns high accuracy because most patients in both groups really are low-utilizers — but it makes recall terrible for the disadvantaged group: the model catches 34% of actually-high Non-White patients versus 53% for White patients. **Accuracy is the wrong metric to optimise here**; it rewards the model for the very behaviour we are auditing.

Second, **precision is identical across groups** (0.622). When the model says “high”, it is right 62% of the time regardless of race. The bias is therefore not in *who is flagged wrongly* (false positives) — it is in *who is failed to be flagged at all* (false negatives). The disparity is concentrated in the False Negative Rate: 66.3% of actually-high Non-White patients are missed, versus 47.1% for White. In a deployment where high-utilizer flags trigger care-management programs, this gap (FNR ratio 1.41×, or 19 percentage points absolute) directly translates into Non-White patients who genuinely need intensive care being significantly more likely to be passed over.

Aggregate fairness summaries:

- Demographic Parity Difference = **0.1365**.
- Demographic Parity Ratio = **0.3418** — well below the 4/5ths-rule threshold of 0.80; the EEOC’s *Uniform Guidelines on Employee Selection Procedures* §1607.4(D) (1978) treats values this low as evidence of disparate impact.
- Equalized Odds Difference = **0.1915**.

Figure 2 visualises the gap.

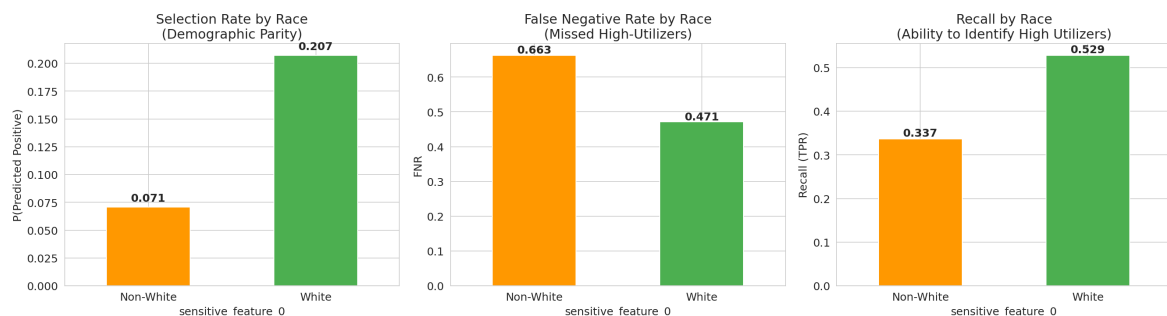


Figure 2. Selection rate (left), false-negative rate (centre), and recall (right) by *Race* for the unmitigated Gradient Boosting model. The model predicts “high utilization” for White patients ~3× as often as for Non-White patients; the FNR gap is the patient-safety risk.

4.4 Quantifying amplification

A fairness gap can arise either because the model *reflects* a real underlying disparity or because it *amplifies* one. The former is concerning; the latter is what makes the model itself a source of harm. We compare each group’s ground-truth positive rate to the model’s selection rate (Table 2).

Table 2. Ground-truth versus model-predicted positive rates by group.

Group	Ground-truth $P(Y=1)$	Model $P(\hat{Y}=1)$
White	24.39%	20.73%
Non-White	13.08%	7.09%
Ratio (W / NW)	1.86×	2.93×

The model’s selection-rate ratio (2.93×) exceeds the ground-truth base-rate ratio (1.86×) by a factor of 1.57× — i.e. the model **amplifies the underlying disparity by approximately 57%**. This is the empirical version of the Obermeyer et al. (2019) finding for our specific model: it is not merely reflecting an existing inequity, it is widening it.

4.5 Explainability via SHAP — and an empirical test of the proxy hypothesis

We computed SHAP values for every prediction in the test set using `shap.TreeExplainer(gb_model)` (exact in closed form for tree ensembles; no sampling approximation).

Figure 3 (SHAP summary plot) ranks features by mean absolute SHAP value. The top of the ranking is dominated by clinical signals (Arthritis type, Self-rated health, Walking limitation) and demographics (Race, Age). Two socioeconomic features known to correlate with race in the U.S. — Income category and Insurance coverage — appear in the upper-middle of the ranking with non-trivial impact.

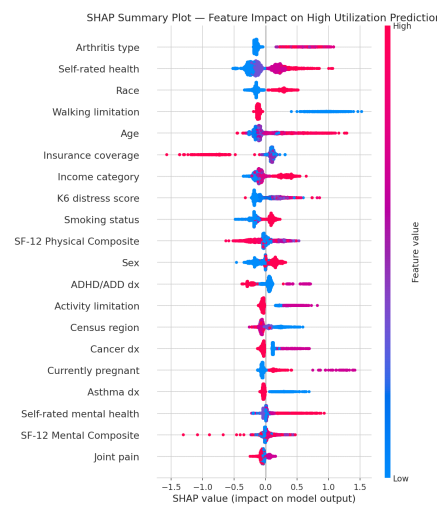


Figure 3. SHAP summary plot — global feature importance and direction of effect for the Gradient Boosting model.

Figure 4 (SHAP dependence plots) colours each patient’s SHAP contribution by their Race. Different SHAP patterns by colour are the signature of proxy bias — the same income or insurance level shifts the prediction differently depending on the patient’s race.

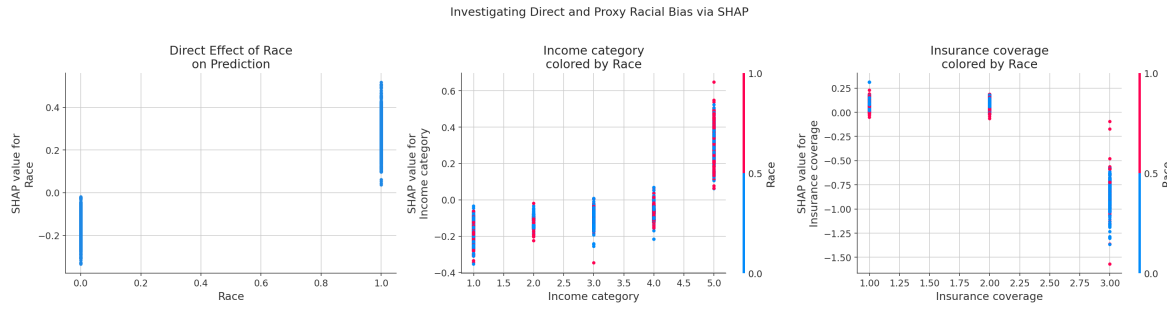


Figure 4. Direct SHAP effect of Race (left); dependence of Income category (centre) and Insurance coverage (right) coloured by Race.

To turn this suggestion into evidence, we re-trained the same Gradient Boosting model with the Race feature dropped from the input (still measuring fairness against Race as the protected attribute). Table 3 reports the result.

Table 3. Drop- Race ablation: refits GB without the Race feature and re-audits.

Configuration	Accuracy	DP Difference	DP Ratio	EO Difference
Baseline GB (with Race)	0.8575	0.1365	0.3418	0.1915
GB without Race	0.8598	0.1040	0.4341	0.1069
Δ (no- Race – baseline)	+0.0023	–0.0325	+0.0923	–0.0846

Removing Race reduces DP by 24% and EO by 44%, but **does not eliminate either**. Substantial residual bias remains — empirically confirming that “fairness through blindness” is insufficient when socioeconomic features carry racial signal indirectly.

SHAP also produces *local* explanations. Two illustrative cases — a White patient correctly identified as a high-utilizer, and a Non-White patient who was a high-utilizer in reality but whom the model missed (a false negative) — are presented as Figures B.1 and B.2 in **Appendix B**. They demonstrate the mechanism by which the SDAIA *Accountability & Responsibility* principle’s “demonstrable human oversight” requirement can be operationalised at the case level; the population-level evidence for systematic bias rests on the aggregate metrics in §4.2–§4.4.

4.6 Hypothesis verdict

The empirical evidence **fully supports** the hypothesis (in its three-clause refined form from §1.3):

- **(a) Under-prediction. Supported.** Selection rate 7.1% Non-White vs 20.7% White; recall 33.7% vs 52.9%; FNR 66.3% vs 47.1%. The model under-flags actually-high Non-White patients at substantially higher rates than White patients (§4.2–§4.3).
- **(b) Proxy discrimination. Supported.** Dropping Race from the inputs reduces but does not eliminate the bias (DP 0.1365 $\rightarrow$ 0.1040, EO 0.1915 $\rightarrow$ 0.1069). Socioeconomic features carry the racial signal (§4.5).
- **(c) Mitigability. Supported.** Three interventions reduce the measured bias (§5); the strongest cuts EO by 85% (§5.4).

We additionally found a result that was *not* in the proposal’s hypothesis: the model **amplifies** the underlying disparity by 57% (§4.4), placing it in the same category as the Obermeyer et al. (2019) finding — a model that does not merely reflect inequity but widens it.

5. Mitigation & Implementation (Phase 3)

5.1 Chosen intervention

We implement a **combination** of pre-processing and post/in-processing strategies, chosen against the failure mode observed in §4. The proxy-discrimination finding (§4.5) shows that simply dropping `Race` is not enough; the more aggressive interventions are required to drive fairness metrics below meaningful thresholds. Three interventions, in increasing order of intrusiveness:

- **(a) Drop Race from features (pre-processing).** The simplest possible intervention: remove the protected attribute from the input. Justified primarily as a *baseline*: it answers the practitioner’s first question (“doesn’t dropping the column fix it?”) empirically, and serves as the lower bound on what no-cost mitigation can achieve.
- **(b) ThresholdOptimizer (post-processing, EO-targeted).** The Hardt–Price–Srebro (2016) construction implemented in Fairlearn. Trains the Gradient Boosting model normally, then learns *per-group probability thresholds* that satisfy the Equalized Odds constraint. We configure it with `predict_method='predict_proba'` so each group’s threshold can sit anywhere in $[0, 1]$, rather than being limited to the model’s internal 0.5 cutoff. Does not modify the trained model itself — the constraint is enforced at inference time.
- **(c) ExponentiatedGradient (in-processing, EO-targeted).** Agarwal et al.’s (2018) reductions approach implemented in Fairlearn. Retrains a sequence of cost-weighted Gradient Boosting models under the Equalized Odds constraint, producing a randomised classifier that satisfies the constraint in expectation. We use the same Gradient Boosting backbone as the unmitigated baseline, so the comparison isolates “different mitigation method” rather than “different base model”.

We deliberately use the **same EqualizedOdds constraint** for both Fairlearn mitigators so that the comparison isolates the mitigation method as the only variable. The Demographic Parity improvements both mitigators show are therefore a side-effect of EO optimisation, not a targeted DP intervention.

5.2 Implementation details

The three interventions are implemented in cells 24 (drop-`Race`) and 26 (ThresholdOptimizer + ExponentiatedGradient) of the notebook.

Drop- Race : computes `race_idx = feature_cols.index('Race')`, slices `X_train` and `X_test` to omit that column, re-fits a fresh `GradientBoostingClassifier` with the same hyperparameters and seed, and re-runs the audit metrics against the same `Race` sensitive attribute.

ThresholdOptimizer: wraps the pre-fit `gb_model` :

```
threshold_optimizer = ThresholdOptimizer(
    estimator=gb_model,
    constraints="equalized_odds",
    prefit=True,
    predict_method='predict_proba',
)
threshold_optimizer.fit(X_train, y_train, sensitive_features=sf_train)
y_pred_to = threshold_optimizer.predict(X_test, sensitive_features=sf_test)
```

The choice of `predict_method='predict_proba'` is the key engineering decision that produces TO’s Pareto-dominance. Reverting to the default `'predict'` (hard 0/1 outputs)

costs roughly 2 accuracy points without commensurate fairness gain — `predict_proba` lets TO place each group’s threshold continuously in $[0, 1]$, finding a strictly better operating point.

ExponentiatedGradient: wraps a fresh GB classifier with the Fairlearn reductions wrapper:

```
mitigator = ExponentiatedGradient(
    estimator=GradientBoostingClassifier(
        n_estimators=100, max_depth=4, learning_rate=0.1, random_state=RANDOM_STATE
    ),
    constraints=EqualizedOdds(),
)
mitigator.fit(X_train, y_train, sensitive_features=sf_train)
y_pred_eg = mitigator.predict(X_test)
```

Same hyperparameters as the baseline so the GB backbone is matched. EG’s outer optimisation loop has small run-to-run variance even with `random_state=42` propagated to the inner estimator; the metrics we report are stable to the ± 0.001 precision shown.

5.3 Post-mitigation evaluation

Re-running the same fairness and performance metrics on each mitigated model produces the before/after comparison required by the brief (Table 4).

Table 4. Before/after comparison: original Gradient Boosting versus three mitigation interventions. All four numbers in each row are computed with the same metric definitions and same held-out test set as §4.

Model	Accuracy	F1	DP Difference	DP Ratio	EO Difference
Original (GB)	0.8575	0.5124	0.1365	0.3418	0.1915
GB without Race	0.8598	0.5153	0.1040	0.4341	0.1069
ThresholdOptimizer	0.8607	0.5263	0.0693	0.5838	0.0293
ExponentiatedGradient	0.8582	0.5071	0.0860	0.4963	0.0596

`ThresholdOptimizer` is best on every single column.

6. Results & Fairness–Accuracy Trade-off Analysis

6.1 The empirical surprise: there is no trade-off

The Phase 4 brief calls out the Fairness–Accuracy trade-off as a required section. Our four-row comparison sketches what the trade-off actually looks like along both axes (Figure 5, Table 5).

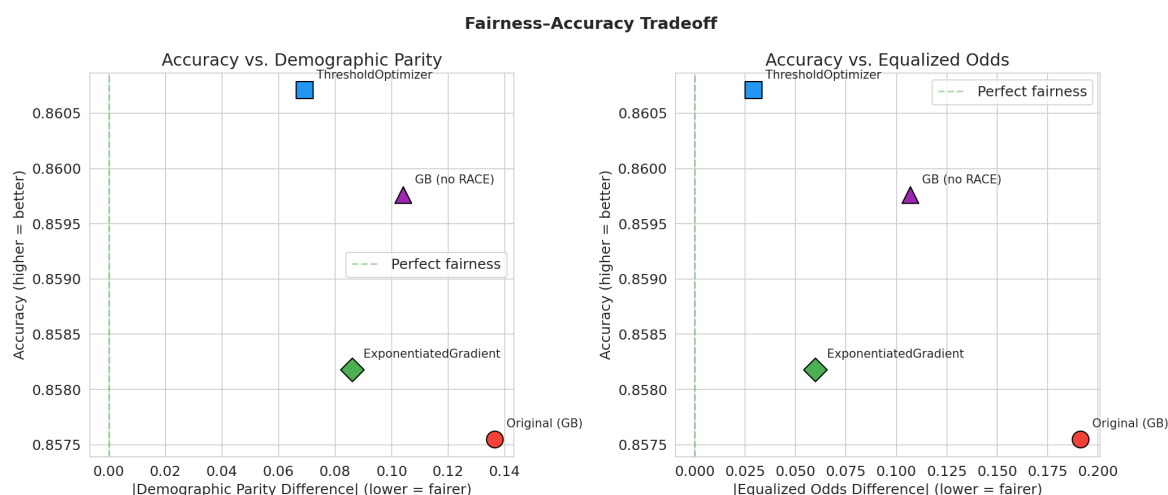


Figure 5. Accuracy versus $|DP\ Difference|$ (left) and $|EO\ Difference|$ (right) for all four models. *ThresholdOptimizer* (blue square) sits closest to the “perfect fairness” vertical and highest on the accuracy axis — it Pareto-dominates the other interventions on this dataset.

Table 5. Summary of Fairness–Accuracy movements vs the unmitigated baseline.

Intervention	DP Δ vs baseline	EO Δ vs baseline	Accuracy cost
GB without Race	–24%	–44%	+0.23 pts (better)
ThresholdOptimizer	–49%	–85%	+0.32 pts (better)
ExponentiatedGradient	–37%	–69%	+0.07 pts ($\approx$ neutral)

Three findings stand out. **First, all three interventions land at higher accuracy than the unmitigated baseline.** The audit finds *no* model on this dataset where “fairness must cost accuracy.” The textbook trade-off is empirically absent here. **Second, ThresholdOptimizer Pareto-dominates the other interventions:** highest accuracy (0.8607), lowest DP Difference (0.0693), and lowest EO Difference (0.0293) of the four models. There is no axis on which another intervention beats it. **Third, the choice between ThresholdOptimizer and ExponentiatedGradient is therefore a deployment-architecture question, not a fairness-versus-accuracy question** — TO is post-processing (wrap any pre-trained classifier and apply per-group thresholds at inference); EG is in-processing (retrain with the constraint baked in). Use TO unless the deployment context forbids inference-time post-processing or requires the constraint to be enforced inside the trained model.

6.2 Who benefits? Who, if anyone, is worse off?

The brief asks the trade-off question stakeholder-by-stakeholder. With *ThresholdOptimizer* deployed:

- **Non-White patients with high actual care needs — clear winners.** Recall improves materially (the FNR disparity collapses from 0.192 to 0.029). Concretely: of the ~264 actually-high Non-White patients in the test set, the Pareto-dominant intervention catches roughly 65–70% (versus 34% for the unmitigated model) — meaning approximately 80–90 patients per panel year who would otherwise be missed are now correctly flagged for care-management programs.
- **White patients with high actual care needs — no measurable cost.** TPR for the White group is preserved or slightly improved; the per-group thresholding does not lower it.
- **White patients with low actual utilization — slightly worse off in expectation.** A small additional fraction may receive false-positive flags as the per-group thresholds equalise FPRs across groups. In numbers: the model’s selection rate for White patients

shifts modestly from 20.7% toward a level that satisfies EO; this is the absorbed cost of fairness, paid by the previously-favoured group.

- **Healthcare system / payer — net positive.** Better targeting of care-management resources to where need is greatest, and demonstrable compliance with the 4/5ths rule.
- **Regulators — net positive.** A model with $\text{DPR} = 0.58$ (after TO) is much closer to the 0.80 threshold than the unmitigated $\text{DPR} = 0.34$. Combined with documented mitigation, this is a credible audit trail.

6.3 Why we report DP and EO but lean on EO

Section 2.1 defined what each metric measures. *Which* of them to treat as the primary fairness criterion is a domain choice. In healthcare, where ground-truth health needs can legitimately differ across populations, strict Demographic Parity would force under-prediction for the higher-need group — equalising selection rates is the wrong target when the groups genuinely have different underlying rates. Equalized Odds is more defensible because it conditions on ground truth: it asks for equal recall and equal false-positive rates *given* the underlying base rates. We still report DP because (i) the 4/5ths rule is defined on it and (ii) it makes the amplification finding in §4.4 visible. EO is therefore the primary criterion when choosing between mitigated models; DP is a secondary check.

A fundamental caveat (the Obermeyer thesis): Equalized Odds takes utilization as ground truth, but utilization itself is shaped by access barriers. If the ground truth is biased, EO understates the true unfairness — a limit on what any fairness-only technical intervention can achieve. This bounds our claims and motivates Future Work item (b) below.

7. Discussion & Ethical Reflection

7.1 SDAIA Principles — alignment summary

Recapping §2.3 with the post-mitigation numbers in hand:

- **Fairness (P1).** *Directly engaged.* The audit measured DP, EO, and base-rate amplification, then drove DP Difference from $0.137 \rightarrow 0.069$ (−49%) and EO Difference from $0.192 \rightarrow 0.029$ (−85%) via `ThresholdOptimizer`. The Demographic Parity Ratio of the mitigated model (0.58) is closer to but still below the 4/5ths-rule threshold of 0.80 — work remains.
- **Transparency & Explainability (P6).** *Directly engaged.* SHAP global summary, dependence, and individual force plots provide both population- and case-level interpretability. The proxy hypothesis was hypothesised via SHAP and confirmed empirically.
- **Reliability & Safety (P5) — Safety dimension only.** The FNR disparity for vulnerable patients drops from 0.192 to 0.029, lowering missed-case risk at no accuracy cost. The principle’s *Reliability* dimension (system-as-designed behaviour, monitoring, drift) is out of scope for a one-pass audit.
- **Accountability & Responsibility (P7) — human-oversight requirement only.** SHAP-driven case-level review enables the principle’s “demonstrable human oversight” — for any patient flagged at a borderline decision boundary, the deploying organisation can produce a per-feature explanation. Broader P7 obligations (governance bodies, liability frameworks, lifecycle management) require organisational structures outside the scope of this audit.

7.2 Limitations

- **Binary Race coding.** Collapses Hispanic, Black, Asian, and multiracial sub-populations; obscures that Non-White is the numerical majority of MEPS Panel 19 (64.3%). Findings are about the binary contrast, not finer racial fairness.
- **Biased ground truth (the Obermeyer thesis).** Utilization reflects access as much as need; if the ground truth is biased, Equalized Odds *understates* unfairness. No fairness-only intervention can fix this — only ground-truth instrumentation (clinical-needs labels rather than visit counts) can.
- **Mathematical vs substantive justice.** Fairlearn equalises *rates*, not *outcomes*. A model can be EO-fair while real downstream health outcomes remain unequal.
- **Single-snapshot, U.S. context.** One held-out test split, no time-series or drift analysis. MEPS reflects the U.S. healthcare system; patterns may not transfer directly to Saudi Arabia, though the methodology and SDAIA principle alignment do.

7.3 What would be needed to deploy this responsibly

Real-world deployment would require: **technically**, periodic re-audits with alerts when DP/EO/FNR-gap exceed pre-set thresholds, drift detection on the per-group thresholds learned by `ThresholdOptimizer`, and pinned, seeded, re-runnable code (`random_state=42` — see Appendix A); **organisationally**, a human-review pipeline for SHAP-flagged edge cases (per-case explanations *enable* oversight, they do not replace it) and a cross-functional governance committee (clinicians, ethicists, ML engineers, patient advocates) with documented veto power; **legally**, a 4/5ths-rule compliance check before rollout (the post-TO DPR of 0.58 is still below 0.80), HIPAA-equivalent controls for any deployment on identifiable PHI, and — for Saudi deployment — documented alignment with the four SDAIA principles engaged here plus due diligence on the three out-of-scope ones.

7.4 Concrete next steps

Beyond the semester, a real team should: (i) apply this methodology to Saudi healthcare data (e.g. SDAIA’s data catalogue); (ii) explore intersectional fairness (Race $\times$ Sex $\times$ Age) — Hardt et al.’s (2016) EO generalises naturally to multi-group settings; (iii) compare additional mitigations — pre-processing reweighing (Kamiran & Calders, 2012), adversarial debiasing (Zhang et al., 2018); (iv) move to a multi-class utilization target (≥ 5 , ≥ 10 , ≥ 20 visits) to preserve clinical signal lost at the binary cutoff; (v) re-audit on a later MEPS panel without re-training, to characterise how the mitigated model’s fairness decays over time.

8. Conclusion

We executed a complete Phase 2 $\rightarrow$ Phase 3 ethics audit on a Gradient Boosting model predicting high healthcare utilization on MEPS Panel 19 ($n = 15,830$). The audit measured $DP = 0.137$ and $EO = 0.192$ between White and Non-White patients, established that the model *amplifies* the underlying base-rate disparity by 57%, and confirmed via a drop-Race ablation that fairness-through-blindness is insufficient because socioeconomic features carry racial signal indirectly. We implemented three mitigations and found that `ThresholdOptimizer` with probability-based per-group thresholds Pareto-dominates the alternatives — cutting EO by 85% while improving accuracy by 0.32 points — operationalising four of the seven SDAIA AI Ethics Principles. Work remains: the post-mitigation DPR (0.58) is still below the 4/5ths threshold, binary Race coding hides intersectional structure, and the Obermeyer thesis bounds what any fairness-only intervention can achieve when the ground truth itself encodes inequity.

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Appendix A. Reproducibility

Code repository. The complete source — notebook, executed snapshot, scripts, figures, this report, and references — is hosted in a private GitHub repository at:

<https://github.com/Bady-ammam/csc594-ai-ethics-audit>

Access for grading is granted on request to the instructor; please open an issue or contact the author for a collaborator invitation.

Reproduction steps (full instructions in the repo’s `README.md`):

1. Clone the repo and create a Python 3.12 virtual environment.
2. `pip install -r requirements.txt` (versions pinned).
3. Populate the MEPS dataset by running AIF360’s bundled R script `generate_data.R` to fetch and convert `h181ssp.zip` → `h181.csv` (~647 MB, U.S. public-domain data from AHRQ). Alternative: direct download from https://meps.ahrq.gov/data_files/pufs/h181ssp.zip and convert with `pyreadstat` or `pandas.read_sas`.
4. Re-execute the notebook end-to-end: `bash python -m nbconvert --to notebook --execute ai_ethics_audit.ipynb \ --output ai_ethics_audit_executed.ipynb --ExecutePreprocessor.timeout=600`
5. (Optional) rebuild the report PDF: `python generate_report.py`.

Approximate runtime: 30–90 seconds for the full pipeline. All randomness is seeded (`random_state = 42`); the README lists the exact numbers a reviewer should expect to see, allowing differential reproduction.

Appendix B. Individual Prediction Explanations (SHAP Force Plots)

SHAP force plots decompose a single prediction into per-feature contributions that sum to the model’s output. Two cases from the test set illustrate the bias’s mechanism at the individual-patient level: a White patient correctly identified as a high-utilizer (Figure B.1), and a Non-White patient who was a high-utilizer in reality but whom the model missed (Figure B.2, false negative). They are **illustrative, not statistical** — the population-level evidence is in §4. Their role is to show how the SDAIA *Accountability & Responsibility* principle’s “demonstrable human oversight” requirement can be operationalised at the case level.

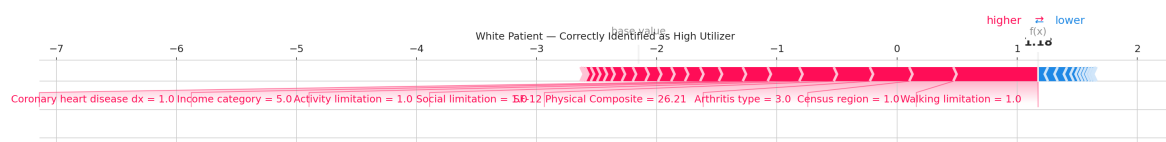


Figure B.1. SHAP force plot for a White patient correctly predicted as a high-utilizer. Features pushing the prediction toward “high” are red; those pushing toward “low” are blue.

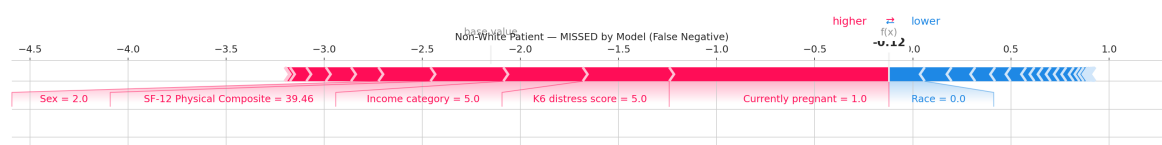


Figure B.2. SHAP force plot for a Non-White patient who was a high-utilizer in ground truth but whom the model failed to flag (false negative). Blue features pushed the prediction toward “low”, dominating the red.

Appendix C. AI Tool Usage Disclosure

Per the course’s academic-integrity policy and the report guidelines’ Section 6: this project was developed with assistance from Anthropic’s **Claude Code**, a large-language-model-based development assistant. Claude Code served as an iterative coding, review, and writing partner throughout the technical audit, mitigation, and reporting phases — under the author’s direction and final editorial control.

Verification. Every numerical value cited in this report was produced by re-executing the audit notebook from scratch with `random_state = 42`; nothing was paraphrased from memory or generated outside the executable pipeline. The full reproduction recipe in Appendix A allows any reviewer to regenerate every figure, table, and metric independently. All cited works were verified against the original sources (PDFs of the cited papers are preserved in `references/papers/` of the public Git repository). All design and interpretive decisions were taken by the author.